

Round 2 (Marketing -> R&D;) Full Documentation (v6)

Multiplayer + 6-Dimension Capability Cards + Budget/Capacity Constraints
March 4, 2026

1 Purpose & What Changed from v5

This document is the single source of truth for Round 2 computation. Hand it to engineering - they can implement without guessing. Hand it to students - they can trace every step.

1.1 v5 -> v6 Change Summary

Area	v5.3	v6
R&D card selection	6 cards from flat list	8-10 cards, 6 dimensions x low/mid/high tier
Data files	feature_card_econ.json	capability_groups_v2.json + compatibility_rules_v2.json
Constraints	None explicit	Hard rules (requires/excludes) + soft (budget/capacity)
Multiplayer	Not covered	4 members, dimension assignment, individual->team merge
Tech Jinnang	Not covered	match_strength modifies dCOGS/risk of matched cards
attach formula	No k_{sub}	$k_{\text{sub}} \times \text{sub_lift}$ added (was in v5.3 appendix, now in main text)
Step 7	Sum 6 cards	Sum 8-10 cards + load + risk floor $\max(0)$ + soft penalty
Step 11 z	No z_{penalty} term	z_{penalty} from budget/capacity overflow added
Anchor calibration	4 x 6-card plans	4 x 8-10-card plans
Market Space	Not in R2 doc	phase4 frozen fields listed for R1->R2 handoff

[v6 NEW] All formulas below are the FINAL v6 versions. Section 11 of v5.3 (appendix-style update notes) has been merged into the main text and deleted as a separate section.

2 Data Objects & Static Configuration

2.1 Runtime Inputs (3 blocks)

- **Round 1 frozen outputs:** grid_id, P, Pmax, f, COGSbase, TAM_units, H, target_gm, market_space_tier, difficulty_tier, vp_scores (C/G/E), jinnang_settle
- **Marketing interview:** radar_scores (6-dim, 1-10), tags [{tag, polarity}], price_sensitive, evi
- **R&D; card selection:** selections [{cap_id, tier}] (8-10 items)

[v6 NEW] Round 1 now freezes market_space_tier and difficulty_tier (computed from market_space_params_v2.json with corrected key format). tech jinnang match_strength is also frozen.

2.2 Static Configuration Files

File	Purpose	Key Fields
capability_groups_v2.json	6 dimension groups, 36 cards, 3 tiers	groups[].capabilities[].tiers.{low,mid,high}.{dCOGS,risk,sub_lift,
compatibility_rules_v2.json	Hard rules + soft constraints	hard_rules[], budget_definition, capacity_definition
grid_priors_v3 (or v4)	12-grid prior weights + recommended cards	grid_weight_prior, recommended_core_tags, recommended_cards
tag_map.json (v2)	Need tags -> dimension mapping	need_tag_to_dim (22 tags), card -> covers mapping
market_space_params_v2.json	Market Space Tier computation	base_size[gridId], elasticity_hint[gridId], thresholds

[v6 NEW] capability_groups_v2.json and compatibility_rules_v2.json replace the old feature_card_econ.json. market_space_params_v2.json keys must use grid_priors format (B2B_Differentiation_Experience), not legacy format (ToB_Differentiation_Elder).

3 Multiplayer Flow (NEW)

[v6 NEW] This entire section is new in v6. v5.3 assumed single-player.

3.1 7-Step UI Flow

Step	Name	Who	Backend	Output
0	R1 Recap	Team	Read phase4 frozen data	Display margin_headroom, market_space, VP scores, jinnang
1	Dimension Assignment	Team	assignDimensions()	4 members -> 6 dimensions (2 per member, overlap OK)
2	Focus Group Interview	Individual	interviewCoach.js	radar_scores, tags, evi per member's assigned dimensions
3	Individual Card Selection	Individual	validateSelections() partial	Each member selects cards for their dimensions only
4	Team Merge	System	mergeTeamSelections()	Combine 4 member selections into unified selections[]
5	Team Discussion	Team	calculate() real-time	Full team can swap cards, all hard rules enforced
6	Final Results	Team	calculate() final	share, units, profit, AI reflection report

3.2 Two-Phase Dependency Enforcement

During individual selection (Step 3), cross-dimension dependencies are shown as **blue info boxes** but NOT enforced - because the member cannot see or control cards in other members' dimensions.

During team phase (Steps 4-5), ALL hard rules (requires/excludes) are fully enforced. The system highlights violations and blocks final submission until resolved.

3.3 Tech Jinnang Application

Tech jinnang cards frozen from Round 1 modify specific R&D; capability cards:

$$dCOGS_{effective} = dCOGS_{base} \times (1 - jinnang_discount)$$

$$risk_{effective} = risk_{base} \times (1 - jinnang_risk_discount)$$

Where jinnang_discount is derived from match_strength (e.g., match_strength=0.87 -> 20% dCOGS reduction). Only applies to cards whose capability matches the tech jinnang's target dimension.

4 Full Computation Chain (Steps 0-12)

The core 12-step chain is UNCHANGED from v5.3 except where marked [v6]. All formulas below are final.

4.1 Step 0: Dimension Set

$D = \{\text{Perception, Mobility, Interaction, Safety, Integration, Operations}\}$

4.2 Step 1: Interview Radar (1-10) -> Normalized s_d

$$s_d = \text{score}_d / 10, s_d \text{ in } [0, 1]$$

4.3 Step 2: s_d -> Interview Posterior Weights

Power transform to emphasize high-scoring dimensions:

$$\begin{aligned} w_{\text{tilde}_d} &= (s_d)^{\text{kappa}} \\ w_{\text{interview}_d} &= w_{\text{tilde}_d} / \text{SUM}(w_{\text{tilde}}) \end{aligned}$$

4.4 Step 3: Prior-Posterior Fusion w^*

$$w_d^* = (1 - \text{lambda} \times \text{evi}) \times w_{\text{grid}_d} + (\text{lambda} \times \text{evi}) \times w_{\text{interview}_d}$$

When $\text{evi}=0$, fully prior; $\text{evi}=1$, maximum lambda shift toward interview.

4.5 Step 4: Severity

$$\text{sev} = \text{SUM}_d(w_d^* \times s_d)$$

4.6 Step 5: Tag Core/Nice Stratification

For each need tag t : $\text{tier}(t) = \text{core}$ if $w_{\text{dim}(t)}^* \geq \text{tau_core}$, else nice.

Importance: $\text{imp}_t = \text{clip}(1 + I(w^* \geq \text{tau1}) + I(w^* \geq \text{tau2}), 1, 3)$

4.7 Step 6: Coverage

$$\text{cover_core} = \text{SUM}(\text{imp} \times \text{covered}) / \text{SUM}(\text{imp}) \text{ [over core tags]}$$

$$\text{cover_nice} = \text{SUM}(\text{imp} \times \text{covered}) / \text{SUM}(\text{imp}) \text{ [over nice tags]}$$

A tag is covered if any selected card's covers set includes it.

4.8 Step 7: R&D Card Aggregation [v6 UPDATED]

[v6 NEW] Now aggregates 8-10 (cap_id , tier) pairs instead of 6 flat cards. Adds load field and risk floor.

$$\text{dCOGS} = \text{SUM}_i(\text{dCOGS}_i), \text{ risk} = \max(0, \text{SUM}_i(\text{risk}_i)), \text{ sub_lift} = \text{SUM}_i(\text{sub_lift}_i)$$

$$\text{load} = \text{SUM}_i(\text{load}_i), \text{ complexity} = \text{SUM}_i(\max(0, \text{dCOGS}_i) / C0)$$

[v6 NEW] risk floor $\max(0)$: negative-risk cards (e.g., quality_system) can offset positive risk, but total never goes below 0.

4.9 Step 7b: Soft Constraint Penalties [v6 NEW]

[v6 NEW] Budget and capacity overflows now produce explicit z-penalties.

From $\text{compatibility_rules_v2.json}$ read budget and capacity definitions:

$\text{budget_cap} = \text{COGSbase} \times \text{budget_multiplier}$ (default: $0.13 \times \text{COGSbase}$)

$\text{positive_dCOGS} = \text{SUM}_i(\max(0, \text{dCOGS}_i))$

If $\text{positive_dCOGS} > \text{budget_cap}$:

$\text{z_penalty_budget} = -\text{slope_budget} \times (\text{positive_dCOGS} - \text{budget_cap}) / \text{budget_cap}$

If $\text{load} > \text{capacity_cap}$:

$\text{z_penalty_cap} = -\text{slope_cap} \times (\text{load} - \text{capacity_cap}) / \text{capacity_cap}$

$\text{risk_add} = \text{risk_per_excess} \times \max(0, \text{load} - \text{capacity_cap})$

Total: $\text{z_penalty} = \text{z_penalty_budget} + \text{z_penalty_cap}$; $\text{risk} += \text{risk_add}$

4.10 Step 8: Weighted Fit

$\text{fit} = \text{SUM}_{\{i \text{ in } S \text{ intersect } R\}}(w_i), \text{ fit in } [0, 1]$

R = recommended capabilities for this grid; w_i from grid_priors recommended_capability_weights.

4.11 Step 9: Price Penalty & WTP

$\text{price_penalty} = \max(0, P/P_{\text{max}} - 1)$

$\text{wtp} = \text{clip}(\text{wtp_base} - \lambda_p \times \text{price_penalty}, 0, 1)$

$\text{wtp_base} = 0.35$ if price_sensitive, else 0.75

4.12 Step 10: Interview Signal I

$I = \text{evi} \times (0.5 \times \text{sev} + 0.35 \times \text{cover_core} + 0.10 \times \text{cover_nice} + 0.20 \times \text{wtp})$

4.13 Step 11: Share & Units [v6 UPDATED]

[v6 NEW] z now includes z_penalty from soft constraints.

```
z = alpha0 + beta_I x I + beta_fit x fit + beta_sub x sub_lift  
- beta_risk x risk - beta_p x price_penalty - beta_cx x complexity + z_penalty  
share = sigmoid(z), M = TAM_units x H, Units = M x share
```

4.14 Step 12: Cost & Profit [v6 UPDATED]

```
COGS = COGSbase + dCOGS  
unit_profit_hw = P x (1 - f) - COGS  
Profit_hw = Units x unit_profit_hw
```

[v6 NEW] attach formula now includes k_sub x sub_lift (was in v5.3 appendix S11.5, now canonical):

```
attach = clip( attach0 + k_core x cover_core + k_nice x cover_nice + k_sub x  
sub_lift - k_risk x risk, 0, 0.9 )  
LTV_sub = attach x S x gm_sub x T  
Profit = Profit_hw + Units x LTV_sub
```

5 Evidence Quality (evi) & Bayesian TargetFit

This section is UNCHANGED from v5.3 Sections 5-6. Key points:

- Single-interview $evi = q \times pT$, where q = reliability score, pT = target-fit posterior
- Reliability (q): deduction method - start at 0.70, deduct 0.10 per missing field / contradiction / constraint violation. Floor 0.30, ceiling 0.90.
- Bayesian pT : prior $O0 = 19$ (95/5), update with $LR_A \times LR_R \times LR_C$ per interview. CPT table unchanged.
- evi enters two paths: (1) weight fusion $w^* = (1-\lambda \cdot evi) \cdot w_{grid} + (\lambda \cdot evi) \cdot w_{interview}$, (2) signal $I = evi \cdot (...)$

See v5.3 Sections 5-6 for full derivation and CPT tables.

6 Parameter Table (v6 Final)

Parameter	Default	Meaning
kappa	1.5	Radar power transform exponent
lambda	0.3	Max prior-posterior fusion shift
tau_core	0.18	Core tag threshold (> 1/6 avg)
tau1, tau2	0.16, 0.22	Importance auto-generation thresholds
alpha	0.4	Recommended card weight mixing (40% count, 60% dim importance)
alpha0	-2.2	Share baseline (sigmoid(-2.2) = 10%)
beta_I	1.2	Interview signal coefficient
beta_fit	2.0	Positioning fit coefficient
beta_sub	1.2	Subscription lift coefficient
beta_risk	1.8	Risk penalty coefficient
beta_p	2.5	Price overrun penalty (strongest)
beta_cx	0.4	Over-engineering penalty
attach0	0.08	Subscription base rate
k_core	0.20	Core coverage -> subscription
k_nice	0.35	Nice coverage -> subscription (higher than k_core by design)
k_sub [v6]	0.25	sub_lift -> subscription (NEW in main formula)
k_risk	0.20	Risk -> subscription erosion
S, gm_sub, T	199, 0.7, 12	Subscription: monthly fee, margin, retention months
C0	1000	Complexity normalization (yuan -> dimensionless)
lambda_p	1.2	WTP price discount intensity
budget_multiplier [v6]	0.13	R&D budget = COGSbase x 0.13
capacity_cap [v6]	24	Max load points before penalty
slope_budget [v6]	-0.15	z_penalty per 100% budget overrun

slope_cap [v6]	-0.10	z_penalty per 100% capacity overrun
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7 Hard Rules & Card Validation (NEW)

[v6 NEW] This section is entirely new in v6.

7.1 Selection Constraints

- Each dimension group: must select 1-3 cards
- Total across all groups: 8-10 cards
- Same-group cards: can coexist (unlike v5 mutual exclusion)

7.2 Hard Dependencies (requires)

Some high-tier cards require a specific card from another dimension. Example: voice_advanced (high) requires perception_base (any tier). Defined in capability_groups_v2.json at the tier level.

7.3 Hard Conflicts (excludes)

Some cards are mutually exclusive across dimensions. Example: edge_compute excludes cloud_heavy. Bidirectional - defined once, enforced both ways.

7.4 Validation Order

1. Check selection constraints (count per group, total 8-10)
2. Check hard rules (requires/excludes) -> return violations[] with repair hints
3. Compute soft penalties (budget/capacity) -> record z_penalty, risk_add
4. Enter economic scoring (Steps 1-12 unchanged)

8 Round 1 -> Round 2 Data Handoff (NEW)

[v6 NEW] Explicit specification of what data flows from R1 phase4 to R2.

8.1 Frozen Fields from phase4

Field	Source	Used In R2
final_grid_id	team.final_grid_id	grid_priors lookup, Fit calculation
final_architecture	team.final_architecture	Display only (Step 0)
target_gm (gm_max)	adjusted.gm_max (after jinnang)	COGSbase derivation, budget_cap
margin_headroom	gmHeadroomTier(gm_max)	Display (Step 0), qualitative label
market_space_tier	computeMarketSpaceTier()	TAM_units modifier, display
difficulty_tier	crowding -> difficulty	Display, qualitative context
vp_scores (C,G,E)	mapVpScores() + LLM	Svol/Swtp computation, display
channel_fee (f)	weighted channel fees	Step 11 z (price), Step 12 profit
COGSbase (cogs_proxy)	base cost after arch/age mods	COGS = COGSbase + dCOGS
Pmax	WTP-derived price ceiling	Step 9 price_penalty
jinnang_tech	settle.tech_adjustments	Modify matched R&D card dCOGS/risk
jinnang_market	settle.market_adjustments	Already applied to gm_max

8.2 Market Space Trade-off

By design, Margin Headroom and Market Space Tier are negatively correlated: high-GM grids (B2B Differentiation) get small base_size -> Space S; low-GM grids (B2C Cost) get large base_size -> Space L. This is achieved through data calibration in market_space_params_v2.json, NOT through formula coupling.

[v6 NEW] Key format in market_space_params must match grid_priors format (B2B_Differentiation_Experience), not legacy format (ToB_Differentiation_Elder). See test_market_space_fix.js for validation.

9 Anchor Calibration

Methodology is UNCHANGED from v5.3 Section 9. Updated for 8-10 card plans:

9.1 Four Anchor Plans (8-10 cards each)

- **Profit-optimal:** High fit + high cover_core + moderate cover_nice + dCOGS within budget. May include 1-2 cost-down cards.
- **Over-engineering:** All high-tier cards. fit very high but dCOGS >> budget, load >> capacity. z_penalty kicks in.
- **Misfit:** Low fit, low cover_core. Cards from wrong dimensions for this grid.
- **Extreme cost-down:** All low-tier + multiple cost-down cards. dCOGS negative but risk high.

9.2 Target Output Ranges

- Profit-optimal: share in [0.15, 0.25], Profit rank #1
- Over-engineering: share may be slightly higher, but Profit LOWER than optimal (budget/capacity penalty)
- Misfit: share in [0.01, 0.05]
- Extreme cost-down: Profit decent but share suppressed by risk

[v6 NEW] Over-engineering now structurally penalized by z_penalty (budget + capacity overflow), not just beta_cx alone.

9.3 Three Parameter Groups to Tune

- Overall level: alpha0 (shift all shares up/down)
- Three hard constraints: beta_fit, beta_risk, beta_p
- Over-engineering penalty: beta_cx + budget slope + capacity slope

10 Implementation Checklist (v6)

#	Task	File(s)	Status
1	Load capability_groups_v2.json + compatibility_rules_v2.json	rdCalculator.js	v6 CODEX ready
2	Implement validateSelections() - hard rules check	rdCalculator.js	v6 CODEX ready
3	Implement computeSoftPenalties() - budget/capacity	rdCalculator.js	v6 CODEX ready
4	Update calculate() - z_penalty in z, k_sub in attach, loadChecking	rdCalculator.js	v6 CODEX ready
5	Run test-rd.js 27 assertions	scripts/test-rd.js	v6 CODEX ready
6	New: assignDimensions() for multiplayer	rdTeamAdapter.js	TO BUILD
7	New: mergeTeamSelections() for team merge	rdTeamAdapter.js	TO BUILD
8	New: applyTechJinnang() modifier	rdTeamAdapter.js	TO BUILD
9	Fix market_space_params key format	market_space_params_v2.json	DONE (test passed)
10	React frontend (7-step flow)	round2_full_v7.jsx	UI prototype done
11	AI interview coach adapt for multiplayer	interviewCoach.js	TO ADAPT
12	AI reflection report generation	reflectionReport.js	TO BUILD

11 Consistency Reminders

- dCOGS_i is ALWAYS relative to LOVOT baseline (additive). Negative = cost-down.
- Cloud service costs go into gm_sub or attach/LTV, never mixed with hardware BOM.
- Software/algorithm investment: either amortized into dCOGS, or one-time R&D deduction (never both).
- risk floor: $\max(0, \text{total_risk})$. Negative-risk cards can offset, but total never below 0.
- budget_cap = COGSbase x budget_multiplier (default 0.13). This is the POSITIVE dCOGS ceiling, not total dCOGS.
- Grid ID format: Always use grid_priors format (B2B_Differentiation_Experience). Never use legacy (ToB_Differentiation_Elder).

[v6 NEW] market_space_params keys were mismatched in production (all base_size lookups fell back to 1). Fixed by re-keying to grid_priors format. Validated by test_market_space_fix.js.

--- End of Document ---